Privacy-Preserving Auction System

1. Introduction

This report explains how four cryptographic components—**Commitments**, **Zero-Knowledge Proofs**, **Secret Sharing**, and **Secure Multi-Party Computation (MPC)**—work together to run a sealed-bid auction where **only the winner and winning bid are revealed**, and all other bids remain private.

2. System Components and Their Roles

2.1 Commitments (Hiding and Binding the Bid)

Commitments allow each bidder to “lock” their bid without revealing it.

Purpose

- Keep the bid completely hidden.
- Prevent the bidder from changing their bid later.

How it integrates

1. Bidder chooses a bid.
2. Creates a commitment to that bid.
3. Sends only the commitment (not the bid itself) to the auction system.

2.2 Zero-Knowledge Range Proofs (Proving Valid Bids)

After committing, the bidder proves their bid is within the allowed range **without showing the bid**.

Purpose

- Confirm the bid is valid (e.g., between 1,000 and 50,000).
- Prevent cheating with negative or extremely large bids.

How it integrates

- Uses the same commitment from Step 1.
- Verifiers can check the bid is valid.
- If the proof fails, the bidder is rejected.

2.3 Secret Sharing (Splitting the Bid Across Servers)

Valid bids are then split into shares and distributed to multiple servers.

Purpose

- No single server can learn any bidder's value.
- Privacy is guaranteed even if one server is compromised.

How it integrates

- The bid is mathematically split into 3 parts.
- Each server receives only one share.
- Shares alone reveal nothing about the bid.

2.4 Secure Multi-Party Computation (MPC) for Comparison

Servers use their shares to jointly compute who has the highest bid.

Purpose

- Determine the winner **without ever reconstructing or revealing any other bids**.

How it integrates

- Servers compare bids using only the shares.
- Only the difference between bids is ever reconstructed (e.g., 8000 – 5000).
- At the end, only the **winning bid** is revealed.

3. Full Auction Flow

Step 1 – Bidder commits to a bid

Each bidder sends a cryptographic commitment that hides their bid.

Step 2 – Bidder proves the bid is valid

They produce a zero-knowledge proof that the bid is within the allowed range.

Step 3 – Valid bids are secret-shared

The bid is split into shares and distributed to three servers.

Step 4 – MPC determines the winner

Servers perform secure comparisons to find the highest bid, learning nothing else.

Step 5 – Only the winner is revealed

The winning bid is reconstructed from shares and announced.
All other bids remain secret forever.

4. Example (Simplified)

Bidder 1: 5,000

- Commitment created
- Validity proof passes
- Bid split into shares

Bidder 2: 8,000

- Same steps
- Bidder 2 also valid

Bidder 3: 999 (Invalid)

- Fails range proof
- Rejected from the auction

MPC result

Servers compare shares of Bidder 1 and Bidder 2.
Only the difference (3000) is ever reconstructed.
Bidder 2 wins.

Final Output

- **Winner:** Bidder 2
- **Winning bid:** 8,000
- All other bids are never revealed.

5. Security Properties

Component	Security Property	What It Protects
Commitments	Hiding	No one sees the bid

Commitments	Binding	Bid cannot be changed
ZK Proofs	Zero-knowledge	Proves validity without revealing bid
Secret Sharing	Privacy	Servers cannot see bids
MPC	Secure computation	Winner found without revealing bids

6. Conclusion

By combining commitments, zero-knowledge proofs, secret sharing, and MPC, the system achieves:

- **Sealed bidding**
- **Bid validity enforcement**
- **Protection of all losing bids**
- **Correct winner selection**
- **Strong privacy guarantees**

This creates a secure, trustworthy, and fully privacy-preserving auction where only the final winning result is public.